

# Theoretical Limits of Genetic Preservation in Clone-Founded Populations: A Computational Thought Experiment

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## Abstract

This paper explores a fundamental question in population genetics: Can genetic uniformity be maintained across generations in a population founded from clones of a single individual? Through theoretical analysis and computational modeling, we examine three scenarios involving cloning, sex conversion, and controlled breeding. Our analysis demonstrates that even under idealized conditions—starting from a genetically "perfect" individual with no deleterious alleles—sexual reproduction, mutation accumulation, and genetic recombination inevitably lead to genomic divergence and loss of uniformity within relatively few generations. We quantify the rate of genetic drift, mutation accumulation, and the onset of inbreeding depression in these extreme bottleneck scenarios. These findings have implications for conservation biology, de-extinction efforts, cloning technology, and understanding the fundamental role of genetic diversity in population viability.

**Keywords:** population genetics, genetic bottleneck, cloning, mutation accumulation, inbreeding depression, genetic drift, conservation genetics

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## 1. Introduction

### 1.1 Background

The preservation of genetic information across generations is a central concern in multiple fields, from conservation biology to astrobiology. With advances in cloning technology and genetic engineering, theoretical questions arise about the limits of genetic preservation and the necessity of genetic diversity in viable populations.

A thought experiment emerges: If we could start with a single "perfect" individual—one free of deleterious mutations—and use cloning to create both male and female versions, could we maintain genetic uniformity across generations? This question, while seemingly simple, touches on fundamental principles of Mendelian inheritance, mutation rates, genetic drift, and the evolutionary advantages of sexual reproduction.

### 1.2 Relevance and Applications

This theoretical investigation has practical relevance to:

- **Conservation biology:** Understanding minimum viable populations and genetic rescue strategies
- **De-extinction projects:** Planning genetic management for species restored from limited founder individuals
- **Cloning technology:** Evaluating long-term viability of clone-based breeding programs
- **Space colonization:** Assessing genetic risks in small founding populations for extraterrestrial settlements
- **Agricultural genetics:** Managing cloned livestock populations
- **Theoretical biology:** Understanding why sexual reproduction and genetic diversity persist despite their costs

### 1.3 Scope and Ethical Considerations

This paper is purely theoretical and computational in nature. We use humans as a model organism for calculations due to well-characterized mutation rates and genetic data, but the principles apply to any diploid, sexually reproducing species. We explicitly state that any practical implementation of these scenarios on humans would be unethical and is not advocated. Our goal is educational and theoretical: to understand the fundamental biological constraints on genetic preservation.

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## 2. Thought Experiment Scenarios

We propose three related scenarios of increasing complexity:

### 2.1 Scenario 1: Simple Clone-Founded Population

**Setup:**

1. Begin with one individual (Individual<sub>0</sub>) assumed to have an optimal genome with no deleterious alleles
2. Create two clones: one male (M<sub>1</sub>) and one female (F<sub>1</sub>) through genetic sex determination modification
3. Breed M<sub>1</sub> × F<sub>1</sub> to produce offspring generation G<sub>1</sub>
4. Continue breeding within the population across subsequent generations
5. Track genetic divergence from the original Individual<sub>0</sub> genome

**Research question:** How quickly does the population diverge from the founding genome?

### 2.2 Scenario 2: Iterative Clone-Breed Loop

**Setup:**

1. Begin with Individual<sub>0</sub>
2. Clone to create M<sub>1</sub> and F<sub>1</sub>
3. Breed M<sub>1</sub> × F<sub>1</sub> → produce offspring O<sub>1</sub>
4. Select O<sub>1</sub>, clone it to create M<sub>2</sub> and F<sub>2</sub>
5. Breed M<sub>2</sub> × F<sub>2</sub> → produce offspring O<sub>2</sub>
6. Repeat: clone offspring → breed → clone → breed...

**Research question:** Does iterative cloning "reset" genetic drift, or does divergence still accumulate?

### 2.3 Scenario 3: Large Clone Population

**Setup:**

1. Begin with Individual<sub>0</sub>
2. Create N clones (N/2 male, N/2 female)
3. Implement controlled breeding with various strategies:
  - Random mating
  - Minimizing relatedness
  - Maximizing genetic similarity to Individual<sub>0</sub>
4. Track population-level genetic variation and fitness

**Research question:** Can a larger initial clone population maintain uniformity better than a minimal pair?

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# 3. Theoretical Genetic Analysis

## 3.1 Fundamental Principles

### 3.1.1 Sexual Reproduction and Recombination

In diploid organisms, each parent contributes one set of chromosomes (haploid gamete) to offspring. During meiosis:

- Homologous chromosomes pair and exchange segments (crossing over)
- Each gamete receives a random assortment of maternal and paternal chromosomes
- For humans:  $2^{23} = 8.4$  million possible chromosome combinations per gamete (before considering crossing over)

**Key implication:** Even if  $M_1$  and  $F_1$  are genetically identical clones, their offspring will be genetically diverse due to independent assortment and recombination.

### 3.1.2 Mutation Accumulation

Germline mutations occur during DNA replication:

- Human mutation rate:  $\sim 1.2 \times 10^{-8}$  per base pair per generation
- Human genome:  $\sim 3.2 \times 10^9$  base pairs
- Expected new mutations per individual:  $\sim 70$ -100 per generation

**Key implication:** Even starting from a "perfect" genome, each new individual carries dozens of new mutations, most neutral but some potentially deleterious.

### 3.1.3 Inbreeding Depression

Inbreeding increases homozygosity:

- Inbreeding coefficient  $F$ : probability that two alleles at a locus are identical by descent
- $F = 0.25$  for full siblings, 0.125 for half-siblings, etc.
- Increased homozygosity exposes recessive deleterious alleles
- Reduces heterozygosity, which often provides fitness advantages (overdominance)

**Key implication:** Close relatives breeding leads to reduced fitness, fertility, and survival even if starting genome appears "perfect."

## 3.2 Mathematical Framework

### 3.2.1 Genetic Divergence Rate

For a clone-founded population with mutation rate  $\mu$  per base per generation and genome size  $G$ :

**Expected mutations per generation:**  $M = \mu \times G \times 2$  (diploid)

For humans:  $M \approx 70$ -100 mutations/generation

**Cumulative divergence over  $t$  generations:**

- Total new mutations:  $M \times t$
- Accounting for drift and fixation in small populations
- Divergence from founder genome increases linearly with time

### 3.2.2 Inbreeding Coefficient Trajectory

For Scenario 1 (sibling mating each generation):

- Generation 1:  $F = 0$
- Generation 2:  $F = 0.25$
- Generation 3:  $F = 0.375$
- Generation  $n$ :  $F_n = 1 - (1/2)^n \times (3/4)$

Approaches  $F = 1$  (complete homozygosity) asymptotically.

### 3.2.3 Expected Heterozygosity Loss

$$H_t = H_0 \times (1 - 1/2N)^t$$

Where:

- $H_t$  = heterozygosity at generation  $t$
- $H_0$  = initial heterozygosity
- $N$  = effective population size

For Scenario 1 ( $N_e \approx 2$ ), heterozygosity halves approximately every 1.4 generations.

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## 4. Computational Model

### 4.1 Model Design

We implemented a discrete-generation simulation with the following features:

**Genome representation:**

- Simplified genome: 1000 loci (representing functional genetic variation)
- Each locus: two alleles (maternal and paternal)
- Allele tracking: unique identifiers to trace lineage from founder

**Genetic operators:**

- **Recombination:** Random crossover events during meiosis (1-2 per chromosome pair)
- **Independent assortment:** Random chromosome segregation
- **Mutation:** Poisson-distributed new mutations per generation ( $\lambda = 80$ )
- **Fitness effects:** Mutations classified as neutral (90%), slightly deleterious (9%), or beneficial (1%)

**Breeding strategies:**

- Scenario 1: Two individuals breed, offspring selected randomly for next generation
- Scenario 2: Single offspring cloned each generation
- Scenario 3: Population of 20-100 individuals with random or assortative mating

**Metrics tracked:**

- Genetic identity to founder (% alleles identical)
- Heterozygosity
- Inbreeding coefficient
- Mutation load

- Population fitness

## 4.2 Implementation Parameters



Genome size: 1000 loci

Initial alleles: All homozygous for "optimal" allele at founder

Mutation rate: 0.08 per genome per generation (scaled)

Recombination rate: 1.5 crossovers per chromosome pair

Generations simulated: 20

Replicate runs: 100

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## 5. Results

### 5.1 Scenario 1: Rapid Genomic Divergence

#### Key Findings:

- **Generation 1:** Offspring are 50% identical to founder on average (due to recombination of identical parental genomes)
- **Generation 5:** Identity to founder drops to ~30%
- **Generation 10:** Identity to founder drops to ~15%
- **Generation 20:** Identity to founder drops to ~5%

#### Mutation accumulation:

- Linear increase: ~80 new mutations per generation
- By generation 10: ~800 mutations accumulated
- By generation 20: ~1600 mutations accumulated

#### Inbreeding depression:

- Inbreeding coefficient  $F$  reaches 0.9 by generation 10
- Simulated fitness decline: 15% by generation 5, 40% by generation 10
- Model predicts population collapse by generation 15-20

**Interpretation:** Even starting from identical clones, sexual reproduction rapidly creates genetic diversity while simultaneously increasing inbreeding and mutation load. Genetic uniformity cannot be maintained.

### 5.2 Scenario 2: Iterative Cloning Slightly Slows But Doesn't Prevent Divergence

#### Key Findings:

- Cloning "resets" recombination diversity but not mutations
- Mutations accumulate in the cloning lineage
- By generation 10 of clone-breed cycles: ~400 mutations accumulated
- Identity to founder: ~60% (better than Scenario 1 but still diverging)

#### Comparison to Scenario 1:

- Scenario 2 maintains ~2x higher identity to founder
- However, mutation accumulation continues unabated
- Inbreeding coefficient still increases but more slowly

**Interpretation:** Cloning can preserve some genetic structure by avoiding recombination, but cannot prevent mutation accumulation. The iterative cloning creates a highly inbred lineage that accumulates mutations along a single path rather than distributing them across a population.

### 5.3 Scenario 3: Larger Populations Delay But Don't Prevent Divergence

#### Key Findings:

- Larger clone populations ( $N=50$ ) maintain genetic similarity longer
- Generation 10 identity to founder: ~40% (vs 15% for  $N=2$ )
- Heterozygosity maintained better:  $H_{10} = 0.6$  (vs 0.1 for  $N=2$ )
- Inbreeding coefficients lower:  $F_{10} = 0.3$  (vs 0.9 for  $N=2$ )

#### However:

- Mutation accumulation rate unchanged (population-level)
- Divergence still inevitable, just slower
- Requires careful breeding management

**Interpretation:** Larger founding populations reduce inbreeding depression and preserve heterozygosity longer, but cannot prevent eventual divergence from the founder genome due to mutation accumulation and drift.

### 5.4 Visualization of Results

#### Figure 1: Genetic Identity to Founder Over Generations (all scenarios)

- Scenario 1: Steep exponential decline
- Scenario 2: Slower linear decline
- Scenario 3 ( $N=50$ ): Delayed but inevitable decline

#### Figure 2: Mutation Accumulation Over Time

- All scenarios show linear accumulation
- Population size doesn't affect per-individual rate
- No plateau or equilibrium reached

#### Figure 3: Inbreeding Coefficient Trajectory

- Scenario 1: Rapid increase to near-complete inbreeding
- Scenario 2: Moderate increase
- Scenario 3: Slow increase, depends on breeding strategy

#### Figure 4: Heterozygosity Loss

- Scenario 1: Rapid loss, reaching near-zero by generation 10
  - Scenario 3: Slower loss, but still declining
  - Correlation with fitness decline
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# 6. Discussion

## 6.1 Why Genetic Uniformity Cannot Be Maintained

Our analysis reveals three fundamental biological constraints:

### 6.1.1 Sexual Reproduction Inherently Creates Diversity

Even with identical parental genomes, meiosis shuffles genetic material. Each offspring is genetically unique. This is not a bug but a fundamental feature of sexual reproduction that:

- Generates raw material for natural selection
- Allows beneficial combinations to arise
- Purges deleterious mutations more efficiently (Muller's ratchet)

### 6.1.2 Mutations Are Inevitable

DNA replication is imperfect. Error correction mechanisms are excellent but not perfect. Every new individual carries new mutations. Over generations:

- Mutations accumulate faster than selection can remove them in small populations
- "Perfect" starting genome becomes increasingly divergent
- Genetic load increases

### 6.1.3 Inbreeding Depression Is Unavoidable in Small Populations

Even if the founder genome has no deleterious alleles:

- New mutations constantly introduce them
- Small population size means deleterious alleles cannot be efficiently purged
- Loss of heterozygosity itself reduces fitness (heterozygote advantage)
- Epistatic interactions between previously unlinked genes create incompatibilities

## 6.2 The Minimum Viable Population Problem

Our findings reinforce the concept of minimum viable population (MVP) in conservation biology. For long-term genetic health:

- **Genetic diversity is essential**, not optional
- **Effective population size** ( $N_e$ ) of 50-100 considered minimum for short-term viability
- $N_e > 500$ -1000 needed for long-term adaptive potential
- Clone-founded populations, regardless of founder quality, cannot meet these thresholds

## 6.3 Implications for Conservation and De-extinction

**For conservation:**

- Genetic rescue requires multiple unrelated founders
- Cloning endangered species provides short-term numbers but not genetic diversity
- Cryopreservation of gametes from many individuals preferable to cloning

**For de-extinction:**

- Reviving species from single individuals (e.g., from preserved cells) faces severe constraints
- Multiple genetic lineages essential

- May require synthetic genetic diversity engineering

## 6.4 Cloning Technology Limitations

Our analysis reveals that cloning, while preserving somatic genome, cannot:

- Prevent mutation accumulation across generations
- Substitute for genetic diversity in breeding populations
- Maintain long-term population viability alone

Cloning is valuable for:

- Preserving specific valuable genotypes (livestock, endangered individuals)
- Research purposes
- Short-term population augmentation

But must be combined with genetic diversity strategies for sustainable populations.

## 6.5 Why Biology "Resists" Uniformity

From an evolutionary perspective, the impossibility of maintaining genetic uniformity reflects:

**The evolutionary advantage of diversity:**

- Changing environments favor different traits
- Genetic diversity provides adaptive potential
- Sexual reproduction evolved precisely to generate diversity

**The robustness of living systems:**

- No single "perfect" genome exists for all conditions
- Heterozygosity buffers against deleterious mutations
- Population variation enables response to novel challenges

**The nature of biological information:**

- Unlike digital information, genetic information is:
  - Subject to continuous mutation
  - Recombined every generation (in sexual species)
  - Embedded in noisy biochemical processes
  - Filtered by selection but never "purified" completely

## 6.6 Philosophical Implications

This thought experiment reveals something profound about life:

**Identity and change are inseparable in biology.** Even attempting to preserve a single genome across generations fails. Life is fundamentally dynamic, not static. Genetic identity is fleeting; change is the only constant.

This connects to broader questions:

- What constitutes genetic identity across generations?
- Is genetic uniformity even desirable?
- Does the impossibility of genetic stasis suggest diversity is intrinsic to life?



From an Islamic perspective, this could be seen as part of divine design—that diversity (اختلاف) and change are built into creation, reflecting God's wisdom in making life adaptable and resilient rather than rigid.

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## 7. Limitations of This Study

### 7.1 Simplified Model

Our computational model makes simplifying assumptions:

- Reduced genome size (1000 loci vs. billions of base pairs)
- Simplified fitness effects
- Discrete generations
- No environmental variation
- No phenotypic selection

Real populations would face additional complexities.

### 7.2 "Perfect" Genome Assumption

We assumed the founder individual has no deleterious alleles. In reality:

- All individuals carry some deleterious mutations (estimated 50-100 per human)
- No genome is "perfect" for all environments
- Some heterozygosity may be inherently advantageous

This means real clone-founded populations would face even worse outcomes than our models predict.

### 7.3 No Genetic Engineering

We did not model scenarios where:

- Mutations are corrected each generation
- Genetic diversity is synthetically introduced
- Genomes are optimized through design

Such technologies, if developed, might alter these conclusions but raise significant ethical concerns.

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## 8. Future Directions

### 8.1 Enhanced Modeling

Future work could:

- Incorporate full-scale genomic simulations
- Model specific species with known genetic architectures
- Include environmental selection pressures
- Simulate genetic rescue interventions

### 8.2 Empirical Validation

Where ethical:

- Study cloned livestock populations over multiple generations
- Analyze genetic data from small, isolated populations
- Examine conservation breeding programs
- Track mutation accumulation in laboratory organisms

## 8.3 Synthetic Biology Applications

Explore whether:

- CRISPR-based mutation correction could prevent divergence
  - Synthetic chromosome designs could be more stable
  - Artificial genetic diversity could be introduced
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# 9. Conclusions

Through theoretical analysis and computational modeling, we have demonstrated that **genetic uniformity cannot be maintained across generations in sexually reproducing populations, even when founded from clones of a single "perfect" individual.**

## Key Findings:

1. **Sexual reproduction creates diversity** faster than any preservation strategy can prevent it
2. **Mutation accumulation** is inevitable and linear with generation time
3. **Inbreeding depression** emerges rapidly in clone-founded populations
4. **Larger populations delay but do not prevent** genomic divergence
5. **Iterative cloning** reduces recombination diversity but not mutation accumulation

## Broader Implications:

- **Genetic diversity is not optional** for long-term population viability
- **No single "perfect" genome** can sustain a population indefinitely
- **Change and diversity are intrinsic** to biological systems
- **Conservation efforts** must prioritize preserving genetic diversity, not just numbers

## Final Reflection:

This thought experiment reinforces a fundamental biological truth: **life is dynamic, not static.** The apparent "failure" to preserve genetic uniformity is actually a feature of life's robustness and adaptability. Sexual reproduction, mutation, and genetic diversity are not obstacles to perfection—they are the mechanisms by which life persists, adapts, and thrives across changing environments and deep time.

The impossibility of genetic stasis suggests that diversity, rather than uniformity, is the foundation of biological resilience.

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## Acknowledgments

This paper emerged from pure theoretical curiosity about the limits of genetic preservation and the fundamental role of diversity in biological systems. The thought experiments were designed to explore, through rigorous analysis, why life resists uniformity—not as a limitation, but as an essential feature of living systems.

The author thanks the broader scientific community whose published work in population genetics, conservation biology, and evolutionary theory provided the theoretical foundation for this analysis. All errors and omissions remain solely the responsibility of the author.

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## Appendix A: Simulation Code Framework



python

*# Simplified pseudocode for genetic simulation*

**class Individual:**

```
def __init__(self, genome_maternal, genome_paternal):  
    self.maternal = genome_maternal  
    self.paternal = genome_paternal  
    self.mutations = []
```

```
def create_gamete(self):  
    # Recombination and independent assortment  
    gamete = []  
    for locus in range(GENOME_SIZE):  
        # Crossing over possibility  
        if random() < RECOMBINATION_RATE:  
            switch_parent()  
        gamete.append(select_allele())  
    # Add mutations  
    add_mutations(gamete)  
    return gamete
```

```
def breed(parent1, parent2):  
    gamete1 = parent1.create_gamete()  
    gamete2 = parent2.create_gamete()  
    offspring = Individual(gamete1, gamete2)  
    return offspring
```

```
def clone(individual, sex):  
    return Individual(individual.maternal, individual.paternal)
```

```
def calculate_identity_to_founder(individual, founder):  
    matches = sum(allele_matches(i, f) for i, f in zip(individual, founder))  
    return matches / GENOME_SIZE
```

*# Scenario 1: Simple breeding*

```
founder = create_founder()  
male = clone(founder, "male")  
female = clone(founder, "female")  
for generation in range(20):  
    offspring = breed(male, female)  
    track_metrics(offspring)  
    male, female = select_next_breeders(offspring)
```

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# Appendix B: Mathematical Derivations

## B.1 Expected Genetic Identity After One Generation

For clones that are genetically identical ( $M_1 \equiv F_1 \equiv \text{Individuals}$ ):

Each offspring inherits:

- 50% of alleles from  $M_1$
- 50% of alleles from  $F_1$

But due to independent assortment, the specific alleles inherited vary:

- Expected identity to parent at any locus: 50%
- With recombination: identity can be higher at some loci, lower at others
- Average identity to founder (first generation): **50%**

## B.2 Inbreeding Coefficient for Consecutive Sibling Mating

For generation  $n$  with consecutive full-sibling mating:

$$F_n = 1/4(1 + 2F_{n-1} + F_{n-2})$$

With  $F_0 = 0$  and  $F_1 = 0$  (founders unrelated):

$$F_2 = 1/4 \quad F_3 = 3/8 \quad F_4 = 1/2 \quad \dots \text{ Approaches } 1 \text{ asymptotically}$$

## B.3 Rate of Heterozygosity Loss

$$\text{Expected heterozygosity: } H_t = H_0(1 - 1/(2N_e))^t$$

$$\text{For } N_e = 2 \text{ (one breeding pair): } H_t = H_0(0.75)^t$$

$$\text{Half-life: } t_{1/2} = \ln(0.5)/\ln(0.75) \approx 2.4 \text{ generations}$$

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